DSN x BCT · DATA & AI SUMMIT · HACKATHON 3.0

TEAM SPRINT PLAN

Task Division by Competency

Who does what · When to do it · How to win

JahsFavour Omoluabi
Data · EDA · Feature Eng · AI

Olanrewaju Muiz
Frontend · API · Backend · AI

Abdulhammed Muh-Awwal
FastAPI · ML · Databases · AI

DEADLINE: 24 MAY 2026
End of Day · No extensions granted

WHAT IS INSIDE THIS DOCUMENT

- Team member roles and daily responsibilities
- 6-day sprint plan — exact tasks per person
- Ownership matrix — who leads every deliverable
- Daily standup format (15-min sync)
- Submission checklist — all 3 deliverables
- 5 tips to maximise your score and win

TEAM ROLES & COMPETENCY MAP

JahsFavour Omoluabi

Data Engineer / Research Lead

Data Analysis · EDA Preprocessing · Feature Eng Naija Context · Paper Lead

Olanrewaju Muiz

AI Prompt Eng / Integration

LLM Prompts · API Wiring Review Gen · Recommender FastAPI + Paper Co-lead

Abdulhammed Muh-Awwal

Backend Infrastructure Lead

FastAPI · ChromaDB Docker · ML Pipeline Repo · README · Infra

SYSTEM ARCHITECTURE — WHO OWNS WHAT

Component	Owner	What it does
Persona Builder	JahsFavour	Extracts tone, rating bias, topics, Naija cues from user history
Task A — Review Gen	JahsFavour + Muiz	LLM prompt simulates user's review text + star rating for unseen items
Task B — Recommender	Muiz	Retrieves candidates from ChromaDB, ranks them, explains WHY each fits
Vector Store (ChromaDB)	Abdulhammed	Indexes all items as embeddings — backbone of Task B retrieval
Cold-Start Handler	Abdulhammed + Muiz	Onboarding Q&A; for new users with no history — worth 25 pts
Nigerian Context Layer	JahsFavour	Injects Naija phrases, cultural items, pidgin into all outputs
FastAPI Endpoints	Abdulhammed + Muiz	/task-a/generate-review and /task-b/recommend — both containerized
Docker + GitHub	Abdulhammed	One-command docker-compose up · clean README · judges must run it
Solution Paper (4-8pp)	JahsFavour + Muiz	Architecture, experiments, ablations, results, Nigerian context section

6-DAY SPRINT PLAN — EXACT TASKS PER PERSON

	JahsFavour	Olanrewaju Muiz	Abdulhammed
Day 1 May 19 *Setup & Foundation*	✓ Download Yelp dataset ✓ Explore + map schema ✓ Profile 500 sample users ✓ Note Naija review patterns	✓ Create GitHub repo ✓ FastAPI skeleton setup ✓ Define request/response schemas ✓ Write basic endpoint stubs	✓ Docker + compose install ✓ Set up ChromaDB locally ✓ Build LLM client wrapper ✓ Test API key connectivity
Day 2 May 20 *Task A User Modeling*	✓ Build persona extractor (JSON) ✓ EDA on rating distributions ✓ Extract Naija cues from text ✓ Validate 10 sample personas	✓ Write review generation prompt ✓ Test 10 review outputs manually ✓ Wire generator to FastAPI ✓ Tune prompt for tone + style	✓ Load Yelp to database ✓ User history retrieval function ✓ Data cleaning pipeline ✓ Feature engineering scripts
Day 3 May 21 *Task B Recommendation*	✓ Prepare item metadata CSV ✓ Generate item embeddings ✓ Index items into ChromaDB ✓ Test similarity search queries	✓ Write recommender agent prompt ✓ Add reasoning to each rec output ✓ Build multi-turn conversation ✓ Test cold-start Q&A; flow	✓ Task B endpoint in FastAPI ✓ Add Amazon dataset (cross-domain) ✓ Cold-start handler function ✓ Optimise DB queries
Day 4 May 22 *Nigerian Context + Integration*	✓ Build Naija phrase dictionary ✓ Cultural items dataset ✓ Inject context into persona flow ✓ Sample 20 Naija review outputs	✓ Connect Task A + B endpoints ✓ End-to-end API test (curl) ✓ Add error handling + logging ✓ Pydantic schema validation	✓ Run docker-compose up — fix bugs ✓ Add sample_data.json for judges ✓ Verify one-command startup ✓ Document env variables
Day 5 May 23 *Metrics & Evaluation*	✓ Run ROUGE-L scoring ✓ Compute BERTScore F1 ✓ Calculate RMSE on ratings ✓ Log all results to CSV	✓ Compute NDCG@10 for Task B ✓ Hit Rate calculation ✓ Test cold-start edge cases ✓ Document all metric results	✓ Record before/after ablations ✓ Test retrieval strategy variants ✓ Prompt variation experiments ✓ Lock final model config
Day 6 May 24 *Paper, Polish & Submit*	✓ Write paper sections 1-4 ✓ Architecture diagram in paper ✓ Experiment tables + Naija section ✓ Final proofread of paper	✓ Write paper sections 5-8 ✓ README final draft + curl examples ✓ Review all API docs ✓ SUBMIT FORM BY 10 PM	✓ Comment all code functions ✓ Remove debug/test code ✓ Clean requirements.txt ✓ Final GitHub push + tag v1.0

SUBMIT BY 10 PM · 24 MAY 2026

Model link + Solution paper + GitHub repo All three in one form before midnight — no extensions

OWNERSHIP MATRIX — WHO LEADS EACH DELIVERABLE

Persona builder	**JahsFavour**	Muiz, Abdulhammed	Feeds all outputs
Review generation prompt	**Muiz**	JahsFavour	ROUGE/BERTScore
Rating accuracy (RMSE)	**JahsFavour**	Abdulhammed	Task A score
Nigerian context layer	**JahsFavour**	Muiz	Bonus + human eval
Vector store (ChromaDB)	**Abdulhammed**	JahsFavour	Task B backbone
Recommender agent	**Muiz**	Abdulhammed	30 pts — NDCG@10
Cold-start handler	**Abdulhammed**	Muiz	25 pts
Cross-domain support	**Abdulhammed**	JahsFavour	Part of 25 pts
Reasoning in rec output	**Muiz**	JahsFavour	20 pts — human eval
Docker + one-command run	**Abdulhammed**	Muiz	10 pts
GitHub repo + README	**Abdulhammed**	Muiz	10 pts
Solution paper	**JahsFavour + Muiz**	Abdulhammed	15 pts — read first

DAILY SYNC — 15-MINUTE STANDUP FORMAT

Question	Who answers	Time
What did I finish yesterday?	All three: J → M → A	3 mins
What am I doing today?	All three: J → M → A	3 mins
What is blocking me right now?	Anyone blocked speaks up	5 mins
Does anyone need my output to proceed?	Check dependencies aloud	2 mins
Are we on track for today's goal?	Team lead (JahsFavour) confirms	2 mins

TASK B SCORING BREAKDOWN — 100 POINTS TOTAL

Criterion	Pts	Owner	Key action to maximise
Ranking quality (NDCG@10)	30	Muiz + Abdulhammed	Retrieve 20+ candidates, re-rank with LLM
Cold-start & cross-domain	25	Abdulhammed + Muiz	Onboarding Q&A; + use Yelp AND Amazon
Contextual relevance (human eval)	20	Muiz + JahsFavour	Nigerian context + reason field in every output
Solution paper	15	JahsFavour + Muiz	Honest, structured, architecture diagram included
Code reproducibility	10	Abdulhammed	docker-compose up works first try, clean README

SUBMISSION CHECKLIST — BEFORE YOU HIT SEND

TASK A — MODEL ENDPOINT

- ■ Persona builder returns valid JSON for any user ID
- ■ Review generator returns rating (float) + review_text (string)
- ■ Nigerian localization active when naija_cues = true in persona
- ■ ROUGE-L and BERTScore computed and recorded
- ■ RMSE on predicted ratings computed and recorded
- ■ /task-a/generate-review endpoint responds with correct JSON

TASK B — RECOMMENDATION ENDPOINT

- ■ Vector store indexed with Yelp + Amazon items (cross-domain)
- ■ Every recommendation includes a 'reason' field explaining WHY
- ■ Cold-start flow works for users with no review history
- ■ Multi-turn works for at least 2 follow-up conversational turns
- ■ /task-b/recommend endpoint responds correctly with ranked list
- ■ NDCG@10 and Hit Rate computed and recorded

SOLUTION PAPER (4-8 PAGES)

- ■ PDF format, between 4 and 8 pages
- ■ Architecture diagram included — clear and labelled
- ■ Experiments section: what you tried and what the numbers show
- ■ Ablation results table included (before vs after improvements)
- ■ Nigerian context section with example review outputs
- ■ All metric numbers match your actual model outputs

CODE REPOSITORY (GITHUB)

- ■ Repo is public (or access shared with judges)
- ■ README has one-command setup: docker-compose up
- ■ README includes example curl calls to both endpoints
- ■ requirements.txt is complete and accurate
- ■ All functions have comments explaining what they do
- ■ sample_data.json included so judges can test immediately

5 THINGS THAT WILL SEPARATE YOU FROM EVERY OTHER TEAM

1. **Nigerian context is your biggest edge**
No other team will do this as deeply as you. Suya spots, Nollywood, pidgin reviews, Detty December. Weave it into persona extraction, review generation, and your paper. It affects Task A human eval, Task B human eval AND the paper impression. This one thing could move you from 3rd to 1st place.

2. **Every recommendation must include a reason field**
Return a 'reason' field with every /recommend response. E.g. 'Recommended because user consistently rates spicy food 4-5 stars and this spot is known for its pepper blend.' Judges doing human eval see an agent that thinks, not just outputs. This is worth 20 points on its own.

3. **Docker must work on the very first try**
docker-compose up should bring everything up with zero manual fixes. If a judge has to debug your setup, you have already lost those 10 reproducibility points. Abdulhammed should test on a fresh directory the evening before submission.

4. **Cross-domain support is free points most teams will miss**
Most teams will use only Yelp. Add Amazon Reviews. A user who reviews food AND products gets cross-domain recommendations. This covers 25 points and is straightforward once ChromaDB is already set up from Task B.

5. **Submit at 10 PM, not at midnight**
Technical issues at 11:55 PM have ended many hackathon dreams. Aim to have everything uploaded and confirmed by 10 PM on 24 May. Use the remaining two hours as buffer, not as coding time. Early submission = calm team = winning mindset.

You have everything you need to win this. The dataset is public, the tools are free, the architecture is clear, and the Nigerian context advantage is yours to take. Execute the sprint, communicate well in the paper, and submit early. Go win it.